

Notable Resources

<https://www.youtube.com/@edmundleung5850>

<https://www.youtube.com/@profrogers6012>

An 2 years out of date full gpu course made by a Purdue Professor

<https://people.inf.ethz.ch/omutlu/>

A completely up to date course on computing by a Onur Mutlu from ETH Zurich (a top European University)

General Resources

<https://ocw.ece.cornell.edu/>

Cornell open course ware resources for CE

<https://ocw.mit.edu/>

The whole MIT open course ware

<https://www.classcentral.com/>

A library to look up courses from different sources

<https://online.stanford.edu/explore>

Stanford open course ware (has the most paid option)

<https://welib.org/>

Database of for everything

<https://www.altera.com/products/development-tools/quartus/compare>

Intel Quartus download options

<https://freecomputerbooks.com/>

A database of books for everything computers

<https://www.youtube.com/@JacobSorber/playlists>

Miscellaneous programming stuff

<https://xilinx-wiki.atlassian.net/wiki/spaces>

Resources for amd stuff (SOCs, Linux, AMD)

<https://swayam.gov.in/>

Indian resource for courses

<https://metacpan.org/>

A resource that focuses on the perl programming language. Can be used for Verilog.

<https://antmicro.com/blog/>

Blogs relating to a lot of hardware stuff

<https://goalkicker.com/>

Free programming books

<https://github.com/drom/awesome-hdl/tree/master>

Resources for programming and hardware languages

<https://github.com/aolofsson?tab=overview&from=2025-12-01&to=2025-12-31>

A list of start ups and programming stuff related to hardware

<https://github.com/sawyermade?tab=repositories>

A github repository with a bunch of different assignments from some guys classes

https://www.youtube.com/@quantum_computing/playlists

Miscellaneous hardware courses on youtube

https://ebruary.net/206292/computer_science/multiprocessor_architectures#1408069

Good general textbook

Compiler

https://web.stanford.edu/class/cs143/?utm_source=chatgpt.com

The stanford resource for compilers.

Linux

<https://0xax.gitbooks.io/linux-insides/content/>

A resource that you can use to start learning about the things you need before you need to understand Linux.

Scala

<https://docs.scala-lang.org/>

General resource for scala

https://twitter.github.io/scala_school/

General resource for scala

C++

<https://github.com/Vitorian/RedditHelp/tree/master>

A github repo full of random stuff meant to help c++ redditors with their homework.

Cybersecurity

<https://pwn.college/dojos>

Miscellaneous resources for cybersecurity

<http://reversing.kr/>

Random cyber challenges

<https://microcorruption.com/>

Random cyber challenges

<https://crackmes.one/>

Random ctfs

Verilog/SysVerilog

https://people.cs.nycu.edu.tw/~ttyeh/course/2024_Spring/CS10014/outline.html

Course description involving verilator with links to labs and other stuff

https://itembedded.com/dhd_list/

Verilator tutorial

<https://zipcpu.com/fpga-hell.html>

Probably the most valuable general purpose verilog/sysverilog resource ever

<https://sutherland-hdl.com/index.html>

Verilog/SysVerilog Tutorial

<https://www.verilogpro.com/all-posts/>

Verilog tutorial

<https://nandland.com/learn-verilog/>

Verilog tutorial

<https://www.maven-silicon.com/free-vlsi-courses/>

A list of courses

<https://verificationguide.com/>

Tutorial for Verilog/SysVerilog/UVM/SystemC/etc.

<https://www.uvmcollab.org/#uvcs>

Tutorial for Verilog, SysVerilog, and UVM

Fpga

<https://github.com/Vitorian/awesome-fpga>

Github repo with resources about fpga stuff.

<https://www.fpga4fun.com/>

Random fpga projects

<https://fpgacpu.ca/>

An extensive list of fpga topics and explanations

<https://projectf.io/>

Extensive fpga stuff

Risc-V

<https://webriscv.dii.unisi.it/>

An entire physical layout of risc-v as you put in different asm instructions

<https://github.com/riscv-boom/riscv-boom>

Full out-of-order RISC-V core. Excellent for learning OoO design.

<https://github.com/chipsalliance/rocket-chip>

In-order RISC-V core generator in Chisel. Industry-influential.

<https://github.com/openhwgroup/cva6>

64-bit Linux-capable RISC-V core.

<https://github.com/cliffordwolf/picorv32>

Tiny, simple RISC-V core — great for beginners.

<https://github.com/SpinalHDL/VexRiscv>

Highly configurable RISC-V core in SpinalHDL.

<https://github.com/chipsalliance/Cores-SweRV>

High-performance dual-issue RISC-V core.

<https://github.com/OpenXiangShan/XiangShan>

State-of-the-art OoO RISC-V core rivaling commercial CPUs.

<https://github.com/black-parrot/black-parrot>

Linux-capable RISC-V multicore architecture.

<https://github.com/lowRISC/ibex>

Small, security-focused RISC-V core.

Cisc

<https://github.com/hughperkins/VeriGPU/tree/main>

Although GPU-oriented, it includes complex pipelines and instruction handling similar to CISC-style decode.

Mips

<https://github.com/jmahler/mips-cpu>

Clean, readable 5-stage pipeline.

<https://github.com/kkdai/mips-cpu>

Good for beginners.

<https://github.com/zhangxiaomian/MIPS-CPU>

Includes pipeline, hazard detection, forwarding.

<https://github.com/yangzhongke/MIPS-CPU>

Adds memory hierarchy concepts.

X86

<https://github.com/jbush001/NyuziProcessor>

Experimental GPGPU-like CPU with SIMD.

https://github.com/MiSTer-devel/ao486_MiSTer

Implements a 486-class microarchitecture.

<https://github.com/darklife/darkriscv>

Minimalist, readable pipeline.

ASIC (Application Specific Integrated Circuit)

<https://www.youtube.com/@ZeroToASICcourse/playlists>

Wide variety of asic stuff

AI accelerator/Tpu

<https://github.com/MatthieuMichon/tiny-tpu/tree/main>

A SysVerilog implementation of a tpu

<https://github.com/nvdla/hw>

Open-source inference accelerator.

<https://github.com/ucb-bar/gemmini>

Systolic array accelerator (TPU-style).

<https://github.com/IBM/AccDNN>

Automatic DNN accelerator generator.

<https://github.com/OpenDRIM/OpenDRIM>

Open deep-learning accelerator architecture.

Gpu

<https://github.com/vortexgpgpu/vortex>

OpenCL-capable GPU architecture.

<https://github.com/VerticalResearchGroup/miaow>

Implements AMD Southern Islands ISA.

<https://github.com/jbush001/NyuziProcessor>

SIMD, multithreaded, GPU-style pipeline.

<https://www.youtube.com/@yifanggpu/playlists>

Full course on gpus

<https://www.youtube.com/@pmpp-book>

Full course on gpus

Microarchitecture

<https://dougallj.github.io/applecpu/icestorm.html>

The firestorm/icestorm microarchitecture overview. It is a reverse engineered thing because apple doesn't want the firestorm/icestorm microarchitecture to be released.

PDK

<https://github.com/google/skywater-pdk>

Includes device models, DRC/LVS decks, standard cells, SRAMs, and documentation.

<https://github.com/google/gf180mcu-pdk>

A 180 nm mixed-signal/MCU-oriented open PDK released in collaboration with Google.

<https://github.com/google/open-source-pdks>

A central index of all open PDKs maintained by Google.

Chisel

<https://github.com/freechipsproject/chisel-bootcamp/tree/master>

Github tutorial for chisel

Mpsoc

<https://github.com/Vitorian/awesome-mpsoc>

Github tutorial for SOC's